PYTHON • DATA SCIENCE • FINANCIAL ANALYTICS

Extracting Stock Data Using Web Scraping

An academic project presentation

Tools: Requests | BeautifulSoup | Pandas

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1. Introduction

Financial analysis relies on accurate and well-organized market data.

Web scraping automates the collection of information presented on webpages.

This project extracts Amazon historical stock information from an HTML table.

The extracted data is organized in a Pandas DataFrame for inspection.

2. Project Objectives

Download webpage content using the Requests library.

Parse HTML elements with BeautifulSoup.

Extract stock-table records and store them in Pandas.

Inspect the first five rows and identify column names.

Retrieve the Open value from the final row and review data quality.

3. Tools and Technologies

Python

Core programming language

Requests

Downloads webpage HTML

BeautifulSoup

Parses HTML and finds elements

Pandas

Structures data in a DataFrame

4. Data Source and Workflow

Source webpage

IBM Developer Skills Network — Amazon stock-data HTML page

Workflow

1. **Request**
2. **Parse HTML**
3. **Extract table**
4. **Build DataFrame**
5. **Inspect**

Source URL: cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/.../amazon_data_webpage.html

5. Retrieving and Parsing HTML

Download webpage content

```
import requests
from bs4 import BeautifulSoup

url = "https://.../amazon_data_webpage.html"
response = requests.get(url, timeout=30)
response.raise_for_status()
html_data = response.text
```

Parse HTML and inspect the title

```
soup = BeautifulSoup(
    html_data, "html.parser"
)

print(soup.title.string)
```

The response text is stored in html_data;
BeautifulSoup creates a navigable HTML structure.

6. Extracting the Stock Table

```
                    table = soup.find("table")
headers = [th.get_text(strip=True)
           for th in table.find_all("th")]

rows = []
for tr in table.find_all("tr"):
    cells = tr.find_all(["td", "th"])
    if cells:
        rows.append([c.get_text(strip=True) for c in cells])

if rows and rows[0] == headers:
    rows = rows[1:]

amazon_data = pd.DataFrame(rows, columns=headers)
```

Key idea

Locate the HTML table.

Read column headings.

Extract each row's cell text.

Create a structured DataFrame.

7. Inspecting the Extracted Data

```
          print(amazon_data.head())
print(amazon_data.columns.tolist())
print(amazon_data["Open"].iloc[-1])

print(amazon_data.shape)
amazon_data.info()
print(amazon_data.isna().sum())
```

What each command reveals

head(): first five records

columns: field names

iloc[-1]: final row's Open value

shape and info(): dataset structure

isna().sum(): missing-value counts

8. Results and Interpretation

The workflow produces a structured DataFrame from HTML table content.

The first five rows and column names can be inspected directly.

The final-row Open value can be retrieved with positional indexing.

Actual values should be copied from the executed notebook; they are not assumed in this presentation.

Verify table ordering and column types before interpreting financial values.

9. Limitations and Data Quality

Scraping code may break if the webpage's HTML structure changes.

The source page may not represent real-time stock prices.

Numeric values may initially be strings and require conversion.

Missing values, duplicates, and formatting should be checked.

Follow the website's access policies and terms of use.

10. Future Improvements

Clean and validate the extracted dataset.

Convert dates and financial columns to suitable data types.

Visualize historical prices and trading volume.

Calculate daily returns and moving averages.

Extend the project to exploratory analysis or time-series modeling.

Consider authorized financial APIs for updated data.

Conclusion

This project demonstrates a practical Python workflow for collecting and structuring financial data from an HTML webpage.

Key takeaway

Requests retrieves the page; BeautifulSoup parses the HTML; Pandas organizes the extracted records for inspection and future analysis.